

Review

Analyses of the MYRRHA spallation loop using the system code ATHLET

Simone Palazzo^{a,*}, Kiril Velkov^{a,*}, Georg Lerchl^a, Katrien Van Tichelen^b^a Gesellschaft für Anlagen und Reaktorsicherheit (GRS), Forschungszentrum, Boltzmannstraße 14, 85748 Garching bei München, Germany^b Belgian Nuclear Research Centre (SCK-CEN), Boeretang 200, 2400 Mol, Belgium

ARTICLE INFO

Article history:

Received 20 November 2012

Received in revised form 13 May 2013

Accepted 13 May 2013

Available online 7 June 2013

Keywords:

ATHLET

RELAP5

MYRRHA

ADS

Spallation loop

LBE

ABSTRACT

One of the most interesting technologies recognized as promising for the future of nuclear energy is the Accelerator Driven System (ADS). The purpose of the ADS is to function as an element of an integrated nuclear power enterprise comprising of conventional and advanced power reactors for energy production, and for reducing the radiotoxicity of the nuclear waste produced by these power reactors before entombment in a geologic repository (Stanculescu, 2000). The use of new types of coolants such as molten lead or lead–bismuth eutectic alloy (LBE) represents another particular feature in the design concept of these reactors, since it permits one to take advantage of higher Heavy Liquid Metals (HLM) boiling temperature compared to water, leading to an improvement in thermal efficiency. Furthermore, it allows the reactor to be operated at a lower pressure, thus reducing the probability of a Loss-Of-Coolant Accident (LOCA) and, consequently, increasing reactor safety. The proposed use of relatively new types of coolants, especially in combination with new fuel types and cladding materials, demands specific attention to the thermal–hydraulics and core mechanics in normal and abnormal conditions. For that reason, a detailed analysis using system codes is necessary.

In this paper, a new version of the ATHLET system code is tested, in which the physical properties of liquid metals like sodium, lead and LBE are implemented. The object of this study is the spallation loop of the MYRRHA facility, in which LBE is circulated by forced convection to remove the heat deposited by a proton beam. A detailed nodalization is set up for performing thermal–hydraulic calculations for both nominal conditions and accident scenarios in order to have a good characterization of the entire loop. The start-up transient has verified the correct removal of the heat generated in the target by the foreseen heat exchanger. During accidental transients, it was noted that the level of LBE in the beam line changes in agreement with preliminary studies on the target device. The pump failure test represents the most dangerous scenario, as the temperature in the target area will reach a very high value within a few seconds after the blockage of the pump impeller. The results obtained have subsequently been compared to the ones achieved with previous numerical simulations which were performed using a version of RELAP5/mod 3.3 system code that was modified at the University of Pisa to account for LBE properties. The comparison has confirmed the capability of the new version of the ATHLET code for the analysis of hydraulic circuits cooled by liquid metals, although both codes use different heat transfer correlations.

© 2013 Elsevier Ltd. All rights reserved.

Abbreviations: ADS, Accelerator Driven System; ATHLET, Analysis of Thermal–hydraulics of Leaks and Transients; BWR, Boiling Water Reactor; CHEOPS, Chemistry Operations; CFD, computational fluid–dynamic; ENEA, Italian National Agency for New Technologies, Energy and Sustainable Economic Development; EURO-TRANS, European Research Programme for the Transmutation of High Level Nuclear Waste in an Accelerator Driven System; GCSM, General Control Simulation Module; GEN IV, Generation IV reactors; GRS, Gesellschaft für Anlagen-und Reaktorsicherheit; HECU, Heat Conduction and Heat Transfer Module; HLM, Heavy Liquid Metal; HTC, Heat Transfer Coefficient; HX, heat exchanger; LBE, lead–bismuth eutectic; LOHS, loss of heat sink; MHD, Magnetohydrodynamic; MYRRHA, Multi-purpose Hybrid Research Reactor for High-tech Applications; NACIE, Natural Circulation Experiment; OECD/NEA, Organization for Economic Co-operation and Development/Nuclear Energy Agency; PHX, primary heat exchanger; R&D, Research and Development; RELAP5, Reactor Excursion and Leak Analysis Program; SCK-CEN, Studiecentrum voor Kernenergie – Centre d’Étude de l’énergie Nucléaire; TDV, time-dependent volume.

* Corresponding author. Tel.: +49 89 32004428.

E-mail address: Kiril.Velkov@grs.de (K. Velkov).

Contents

1. Introduction	275
2. MYRRHA/XT-ADS reactor	276
2.1. Spallation loop	276
2.2. Main components characteristics	277
3. Spallation loop nodalizations	277
3.1. Implementation of the LBE properties in the system code ATHLET	279
3.2. Modeling of heat transfer	280
4. Transients simulation	281
4.1. Start-up and steady state analysis	281
4.2. Accidental scenarios	283
5. Summary and conclusions	285
References	285

1. Introduction

The future of the nuclear energy is strictly related to the development of Generation IV reactors by many research centers and scientific organizations (NEA/OECD, 2002). The main goals of these reactors are sustainability, safety, economic competitiveness and security against weapons proliferation.

The current debate mainly concerns the need to decrease the amount of highly radioactive long-lived material generated during power production. A possible route to reduce the hazard of nuclear wastes is represented by transmutation process in an Accelerator Driven System (ADS). This option combines a particle accelerator with a sub-critical reactor core into a single system that has the potential to efficiently generate electricity from nuclear fission and/or transmute the long-lived waste materials. The protons are delivered into a spallation target to produce the neutrons by the spallation reaction, keeping the nuclear chain reaction in sub-critical core going (Mansani et al., 2011).

As a large amount of heat is deposited into the target after starting the spallation reaction, Heavy Liquid Metal (HLM) has been considered to be both the primary coolant and spallation target material. Solid targets have been developed in the past, but have reached their physical limits in terms of long-time performance and coolability for the conditions necessary in an ADS.

In order to demonstrate the technical feasibility of transmutation in an ADS reactor, the integrated project of the European community EUROTRANS (European Research Program for the Transmutation of high-level Nuclear Waste in an Accelerator Driven System) was started in April 2005. The scope of this project was to design an advanced design of an approximately 50–100 MW_{th} power experimental facility that uses lead–bismuth eutectic alloy (LBE) as the working fluid. This facility was also considered to be the reference test facility for the development of ADS technology (Abderrahim, 2011). At the end of this EUROTRANS project, it was concluded that the advanced XT-ADS design has complied with the main project requirements. Nevertheless, additional R&D studies and qualifications were requested before advancing to the writing of tender specifications. Till date, different spallation target geometries for ADS have been investigated. One of the most interesting configurations is the windowless target, in which no physical separation between the beam tube and the HLM target is present. The stability of the target free surface within this complex device has a crucial effect on its thermal behavior and consequently on its safety. At the same time, the thermal-hydraulics of the target must ensure that the heat deposited into the HLM during the spallation reaction can be safely removed (Batta and Class, 2009). For these reasons, a detailed analysis of the spallation

loop behavior during operational and accidental transients using a system code is mandatory.

In this paper, a new version of the system code ATHLET (Analysis of Thermal–Hydraulics of Leaks and Transients), developed by the German Gesellschaft für Anlagen-und Reaktorsicherheit (GRS), was used for the first time, in which the properties of liquid metals as lead, sodium and LBE have been implemented. The object of the current study is the spallation loop in the MYRRHA/XT-ADS (Multi-purpose Hybrid Research Reactor for High-tech Application/ eXperimental Transmutation in an Accelerator Driven System) facility concept, which was developed by the Belgian nuclear research institute SCK-CEN. MYRRHA reactor design is considered the starting point for the development of lead fast reactor technology (Guertin et al., 2011). After a short description of the MYRRHA/XT-ADS, the main characteristics of the spallation loop are described along with the nodalization of this device.

The results obtained were compared with previous simulation results from a modified version of the RELAP5/mod 3.3 system code, in which the lead and LBE properties have been implemented at University of Pisa (Bianchi et al., 2001). These comparisons create the prerequisites to comprehend the real capability of the new

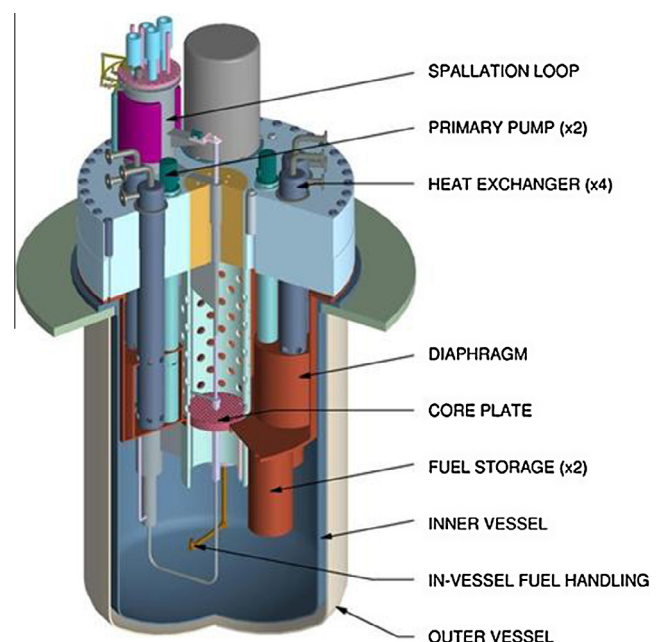


Fig. 1. Simplified representation of the MYRRHA XT/ADS primary cooling system (Class et al., 2011).

ATHLET code version to achieve reasonable values of the main thermal–hydraulic parameters for the new reactors cooled by HLM.

2. MYRRHA/XT-ADS reactor

MYRRHA is an experimental reactor which is planned to be built at the Belgian Nuclear Research Centre SCK-CEN. It consists of a sub-critical nuclear reactor core ($k_{\text{eff}} \approx 0.95$) coupled to an external neutron source. In this external source, neutrons are produced by the spallation reaction, caused by a high intensity proton beam (3.0 mA and 600 MeV) which impinges on a HLM target (LBE). This facility can be used for material testing and fuel research for the future and current generation of nuclear power plant, but also for the research in transmutation of nuclear spent fuel and medical isotope reactor (Maes, 2006).

The reactor is a pool-type reactor, which is filled with liquid LBE. This alloy was chosen as coolant due to the desired high fast flux and the high power density. The melting and boiling points of LBE are 129 and 1670 °C respectively. The pool system is able to contain the primary coolant within the main vessel, in which the main components are inserted from the top (see Fig. 1).

In the reactor main vessel, the hot and cold pools are separated by a cylindrical inner vessel (diaphragm). LBE circulates by means of two mechanical pumps and the power generated by the core is removed by four (2×2) LBE/water/steam primary heat exchangers (PHX). The four PHX are inserted in two circuits, each including two PHX and one pump. The two pumps are axial flow pumps, which run the LBE from the hot pool at the level of the PHX outlet window to the cold plenum. The elevation of the heat exchanger has been designed to achieve natural circulation capabilities.

For the MYRRHA/XT-ADS, the core power has been fixed to 85 MW_{th} as reference value in accordance with the actual reactor design evolution, 74.5 MW_{th} of which are allocated to fission power. This core reaches a total neutron flux of 3.3×10^{15} (n/cm² s) and a fast (>0.75 MeV) neutron flux of 10^{15} (n/cm² s).

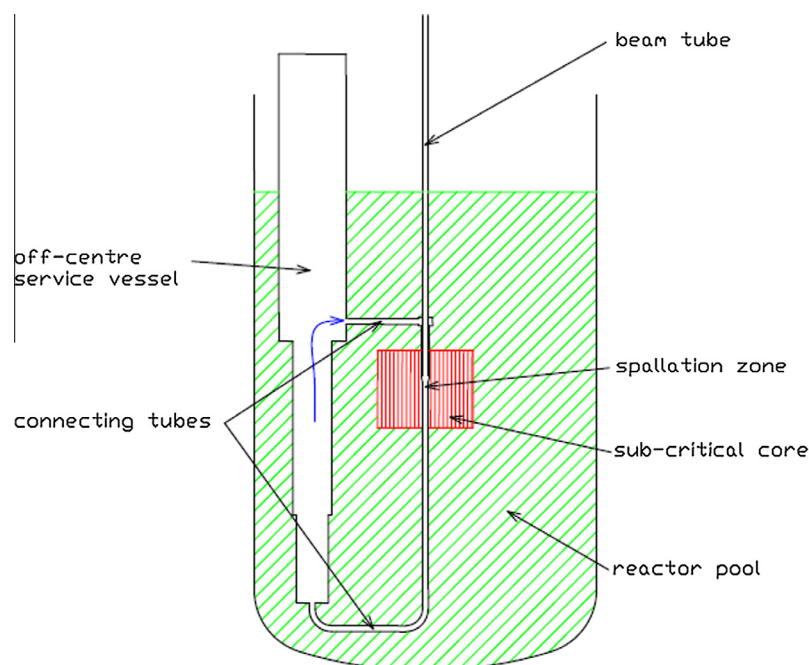


Fig. 2. General layout of the MYRRHA reactor vessel showing how the spallation target is split up in a center and an off-center part and how the two parts are connected (Guertin et al., 2011).

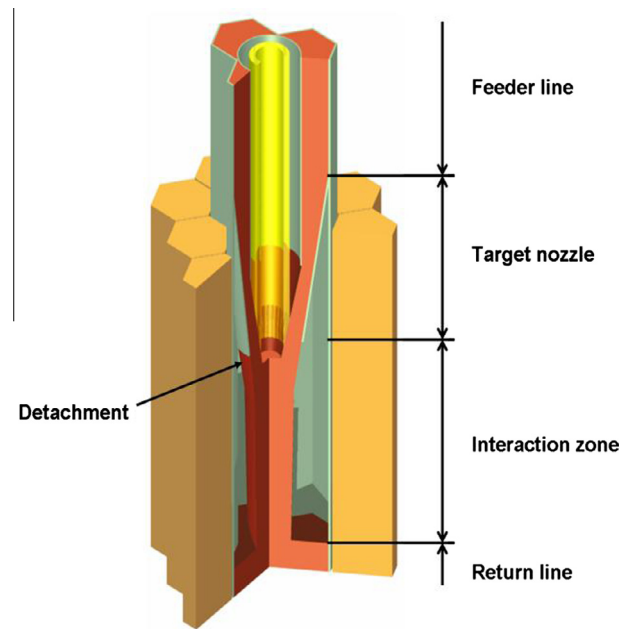


Fig. 3. A cut-out view of the spallation target area with a detached flow design. The LBE flow is indicated in red, the proton beam in yellow and the fuel assemblies in orange (Class et al., 2011). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

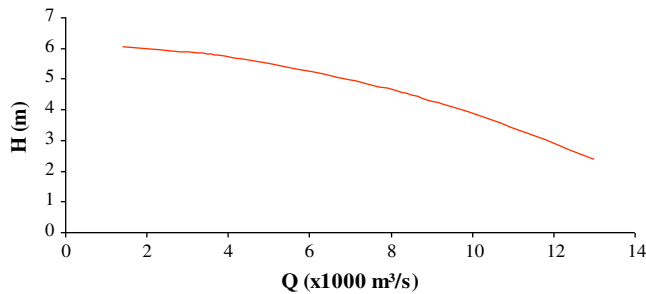
2.1. Spallation loop

The spallation loop represents one of the most innovative components of the MYRRHA/XT-ADS. It can be considered as a physical and functional interface between a particle accelerator and a sub-critical reactor core (Class et al., 2011). The spallation loop consists basically of a big service vessel filled with LBE, which is located beside the sub-critical core. From this main vessel, LBE flows into the target zone, where it interacts with a 600 MeV proton beam to pro-

Table 1

Technical data and dimensions of the spallation loop main pump.

Nominal flow rate	0.013 m ³ /s
Minimal suction head	0.7 bar
Rotation speed	1450 rpm
D_{housing}	400 mm
L_{housing}	400 mm

**Fig. 4.** Characteristic pump curve (Dierckx et al., 2009).

duce neutrons by the spallation reaction. Two horizontal tubes link the central vertical tube to the main service vessel (see Fig. 2).

The service vessel contains the main circulation pump, driven by a hydraulic pump-turbine with electric motor, an auxiliary system for the control of the O₂ content of the coolant, the measure instrumentation and the vacuum system. The whole spallation vessel is kept under vacuum to avoid the contamination of the beam line by gas released at the interface between the spallation target and the proton beam. Only the spallation zone is positioned at the centre of the core, while all other components are moved towards the edge of the reactor pool, where radiation levels are lower and more space is available.

In the MYRRHA/XT-ADS design, three assemblies at the centre of the core are removed to create enough space for the spallation target. In the upper region of the target zone, the available cross section needs to be distributed between beam tube and LBE feeder line, whereas the lower part is completely available to allow the liquid LBE flowing out of the target. The LBE contained in the feed tank flows by gravity in the annular tube surrounding the proton beam line and forms a fluid free falling jet. ALBE conical free surface creates just underneath the end of the beam tube (see Fig. 3). The proton beam impinging on the spallation target generates a power of approximately 1.25 MW and about 80% is converted into heat in the fluid (Maes, 2006). The heat is extracted by a heat exchanger (HX) installed in the lower part of the service vessel. The heat sink of this HX is the primary coolant flowing in the lower plenum of the reactor pool.

According to the last version of the target design, a forced LBE flow detachment from the target wall in the nozzle region is expected. Due to this nozzle shape, an additional free surface is formed in the return line (see Fig. 3). This target design is chosen to improve the shape and stability of the conical free surface, avoiding, in particular, the generation of a large and fluctuating recirculation zone along the central axis of the target.

The maintenance of a stable flow path within the target is mainly determined by the total mass of LBE flowing into the target region. The LBE flowing out of the target does not influence the shape and stability of the target zone as long as the additional LBE free surface in the return line of the target is not completely filled with fluid. Such extra LBE free surface (buffer zone) is used to handle the effect of sudden changes in beam power which result in sudden change of fluid temperature and properties. The annular region above the target zone is equipped with a drag enhancer.

Table 2

Dimensions of the HX tube bundles.

Tube bundle	
Array type	Hexagonal
Number of tubes	349
Outer diameter	15.875 mm
Thickness	1.651 mm
Length	704 mm
Pitch	1.5 D_{out}
Vessel side	
Shroud diameter	483 mm
Flow rate	133 kg/s

This device makes sure the friction losses in the feeder line compensate the energy gain of the fluid due to the gravity (Class et al., 2011).

2.2. Main components characteristics

As already mentioned above, the spallation loop consists of a lot of devices and measurement instrumentations, which guarantee its correct hydraulic behavior during the operational conditions. In this section it will be briefly introduced some technical features and characteristic parameters of the most important devices composing the spallation loop, namely the main pump, the HX and the off-centre service vessel.

Regarding the main pump, the two different options that have been researched are the centrifugal and magneto-hydrodynamic (MHD) pump. The centrifugal option is taken into account in this phase of the study and its main parameters are summarized below in Table 1 and Fig. 4.

The HX designed to remove the heat generated in the spallation target is of the straight tube, counter-current type. Both primary and secondary sides of the HX use LBE as cooling fluid. The dimensions of the tube bundles and of the outer shroud are specified in Table 2.

The service vessel has two essential functions. It must contain the thermal expansion of the LBE during the start-up phase and minimize the emanation of the spallation products into the beam line. A vacuum/cover gas system that is responsible for the confinement of volatile spallation products is installed near the top of the vessel. The pressure directly above the spallation target should be around 10^{−3}–10^{−4} mbar to guarantee compatibility with the vacuum of the proton beam line.

3. Spallation loop nodalizations

The ATHLET nodalization of the spallation loop has been developed mainly on the base of the reference data showed in the previous section. Each component has been implemented in a very detailed way. In the present section, the adopted nodalization is showed (see Fig. 5). The LBE flows in a clockwise direction within the primary loop as well as in the secondary side of the HX.

As regards the primary loop, the complexity of the fluid shape in the target zone has entailed further effort to simulate its thermal-hydraulic behavior as realistically as possible.

Because of the impossibility to generate a control volume with moving wall able to predict the radial dimension of the free falling jet during the transient simulations, fixed measurements have been employed, based on theoretical calculations performed on the last design version of the target. This important simplification has caused the absence of the additional free surface which acts as a buffer during the transients, damping the variations of the free surface in the beam tube.

A minor branching between the first control volume of TAR-GET_ZONE object and the last one of LEVEL_PIPE object has been

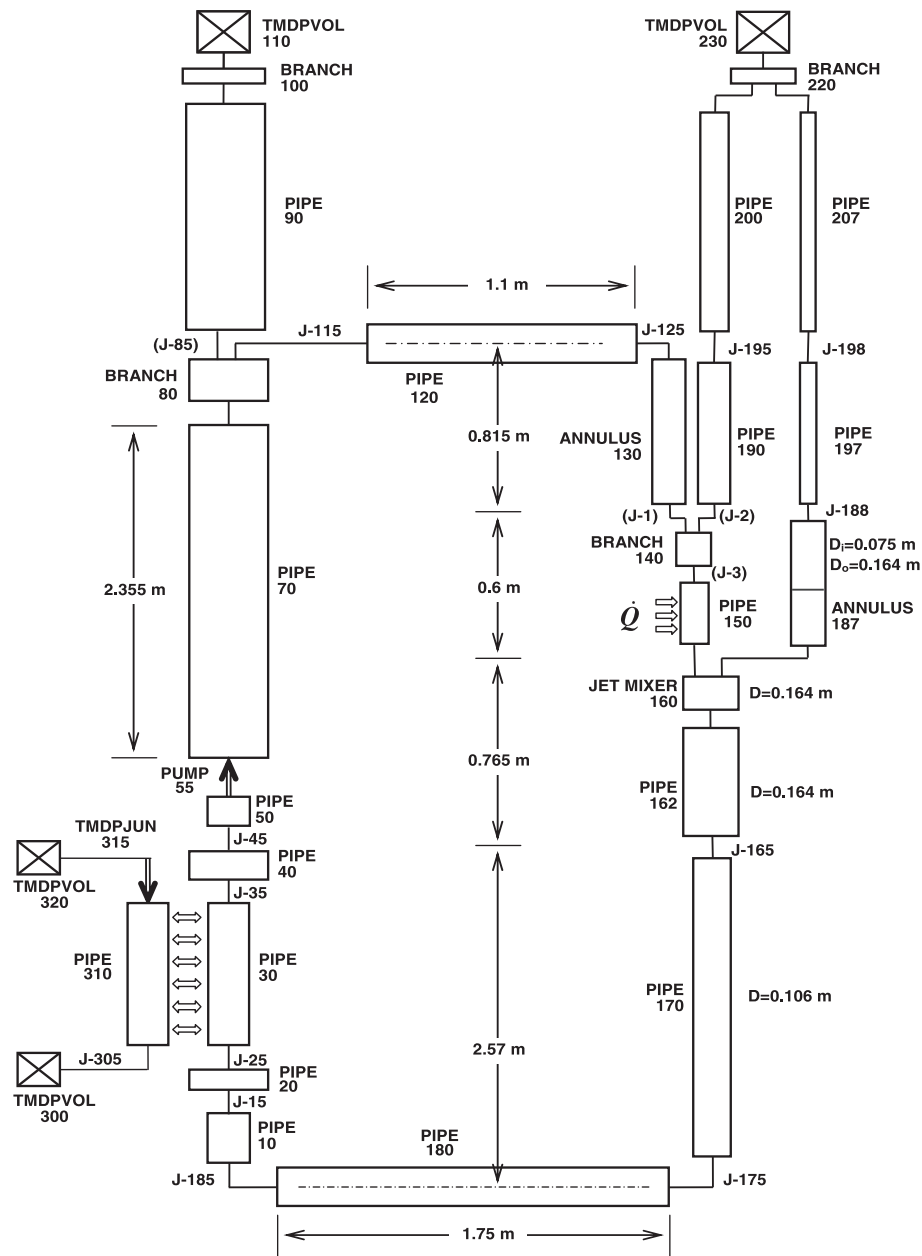


Fig. 6. RELAP5 nodalization of the spallation circuit (Dierckx et al., 2009).

file and linked to the circuit by means of a “jet mixer” component at the lower side of ANNULUS 187. Within these dummy components, a liquid–gas mixture level has been considered using argon as non-condensable cover gas in the upper part of the circuit. Such workaround has also permitted the modeling of the additional free surface (buffer zone), which is placed in the return line of the target zone, just underneath the conical free surface.

The aim of this configuration was to better investigate the behavior of the two different LBE free surfaces in terms of liquid level variation during the accidental transients.

The hydrodynamic boundary conditions for a closed loop are modeled with the time-dependent volumes (TDVs), in which constant values of pressure and temperature for liquid or gas phase can be set up. In the current nodalization, two TDVs are connected at the top of the circuit through a junction, in which a constant pressure of 1 bar has been fixed as reference state value. This simplification was necessary, since the RELAP system code suffers

from numerical issues when the pressure inside the hydraulic system reaches very low values.

3.1. Implementation of the LBE properties in the system code ATHLET

The thermodynamic properties and transport quantities for LBE were obtained from the OECD/NEA report released in 2007 (OECD/NEA, 2007). In this document, correlations proposed by different authors are compared and the most promising is suggested for implementation in simulation programs. The liquid metal properties are given as functions of the liquid metal temperature. Since the partial derivative for pressure is important for ATHLET, small modifications were made to introduce pressure dependence in the liquid metal density correlation. Table 3 shows the OECD/NEA properties of LBE, which were implemented in the system code ATHLET.

Table 3
LBE properties implemented in ATHLET.

Composition	Pb 44.5–Bi 55.5	wt%
Molar mass	208.18	g mol ⁻¹
Latent heat of melting	397.7	K
Boiling temperature	3.86E+04	J kg ⁻¹
Latent heat of boiling (Q boil)	1943	K
Critical temperature	8.54E+05	J kg ⁻¹
Critical pressure	4890	Pa
Critical density	88.0E+06	kg m ⁻³
Saturation vapor pressure	2170	Pa
Density	$\rho_{\text{sat}} = 1.11 \times \exp\left(-\frac{22.552}{T}\right)$	kg m ⁻³
Specific heat capacity	$\rho = 11096 - 1.3236T$	J kg ⁻¹ K
Sound velocity	$C_p = 159 - 2.72 \times 10^{-2}T + 7.12 \times 10^{-6}T^2$	m s ⁻¹
Elastic modulus (adiabatic)	$u_{\text{sound}} = 1773 + 0.1049T - 2873 \times 10^{-4}T^2$	Pa
Thermal conductivity	$\lambda = 3.61 + 1.517 \times 10^{-2}T - 1.741 \times 10^{-6}T^2$	W m ⁻¹ K ⁻¹
Dynamic viscosity	$\eta = 4.94 \times 10^{-4} \times \exp\left(\frac{754.1}{T}\right)$	Pa s
Surface tension	$\sigma = 0.437 - 6.6 \times 10^{-5}T$	N m ⁻¹
Thermal expansion	$\beta_p = \frac{1}{(8383.2 - T)}$	K ⁻¹

It should be mentioned that lead, LBE and sodium are considered in ATHLET as subcooled and subcritical single phase liquid. The state of saturation and boiling cannot be simulated with the current ATHLET version. The temperature range for the new coolants starts from the melting point and ends 10 K below the boiling temperature. The liquid metal pressure ranges from 10 Pa to 10 MPa.

3.2. Modeling of heat transfer

The liquid metal coolants have significantly higher heat conductivity and density (sodium is an exception) and lower specific heat capacity than water. As a result the dimensionless Prandtl number Pr is significantly lower than 1. It represents the ratio of momentum diffusion to the thermal transport in the fluid. Since liquid metals and water differ notably in the convective heat transport, the existing heat transfer correlations (HTCs) for water should not be used for the simulation of liquid metals. To overcome this restriction, research activities have been carried out to select appropriate correlations for the HECU (Heat Conduction and Heat Transfer) module in ATHLET.

In OECD/NEA report (2007), a large number of HTCs available from the literature are present. These were derived for different geometries and flow conditions. Cheng and Tak (2006) concentrated on turbulent heat transfer to LBE flows in circular tubes for nuclear applications and compared existing LBE HTCs with experimental data and computational fluid-dynamic (CFD) calculations. They suggested the following correlation for the estimation of the Nusselt number Nu :

$$Nu = A + 0.025Pe^{0.8} \quad (2)$$

where $A = 4.5$ if $Pe < 1000$ $A = 5.4 - 9 \times 10^{-4}Pe$ if $1000 \leq Pe \leq 2000$ $A = 3.6$ if $Pe > 2000$

For bundle geometries, Pfrang and Struwe (2007) compared different HTCs for liquid metals. The correlation derived by Mikityuk (2009) was implemented in ATHLET:

$$Nu = 0.047(1 - \exp(-3.8((p/D) - 1)))(Pe^{0.77} + 250) \quad (3)$$

where $30 \leq Pe \leq 5000$ and $1.1 \leq p/D \leq 1.95$

p = fuel rod pitch [m], D = hydraulic diameter [m]

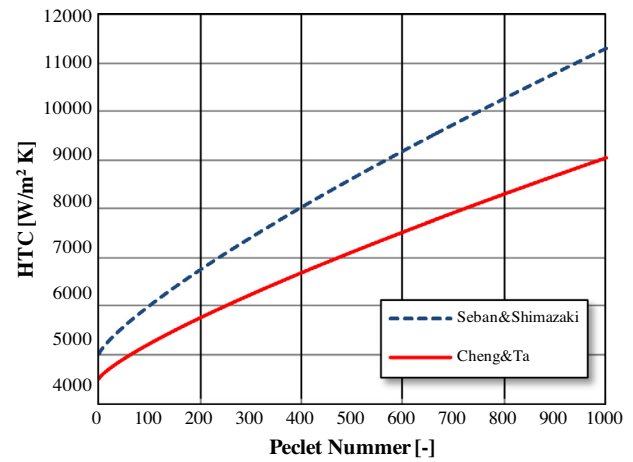


Fig. 7. Comparison of HTC correlations for inner side of tube evaluated at $T_{\text{ave}} = 380$ °C.

This correlation is derived for the tube bundle disposed in a triangular or square lattice and it should be used with care for rod bundles with wire wraps. Furthermore, it represents the mean value of other correlations and it has not been proved that it will provide the best approximation in all situations. In some particular cases, other correlations might give even better results. Therefore, special attention will be paid to the performance of the selected HTCs in the forthcoming validation activities.

For what concerns the RELAP5 system code, the wall-to-fluid correlations implemented in the current version for the calculation of the HTC are quite different with respect to those proposed in the ATHLET code. The Seban and Shimazaki (1951) correlation is suitable for turbulent flow in tubes with constant wall temperature and it is used for the calculation of the Nusselt number inside the HX tubes:

$$Nu = 5.0 + 0.025Pe^{0.8} \quad (4)$$

Regarding the correlation of the HTC calculation for the tube bundle disposed in a triangular lattice, the Kirillov and Ushakov (2001) correlation is taken into account:

$$Nu = 7.55(p/D) - 20/(p/D)^{13} + (0.041/(p/D)^2)Pe^{(0.56+0.19(p/D))} \quad (5)$$

where $1 \leq Pe \leq 4000$ and $1.2 \leq p/D \leq 2.0$

The employment of different HTC correlations in ATHLET and RELAP5 codes causes slight mismatch in the evaluation of Nusselt

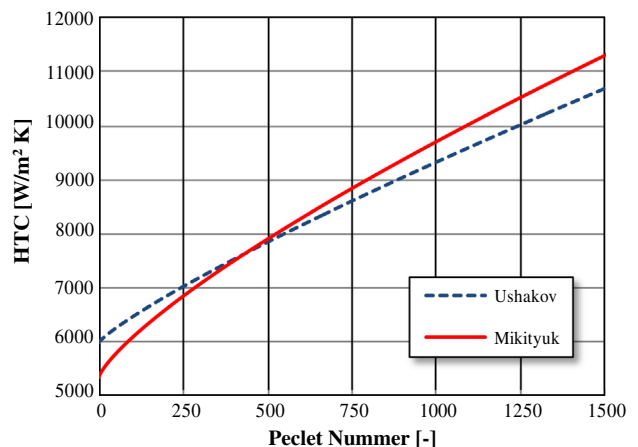


Fig. 8. Comparison of HTC correlations for tube bundles evaluated at $T_{\text{ave}} = 380$ °C and $p/D = 1.5$.

numbers and, consequently, in the temperature values calculated through the target and the HX. All other parameters being equal, some discrepancy in the main thermal–hydraulic variables, such as LBE mass flow and free surface liquid levels, can be therefore pointed out between the two system codes during the steady state and accidental transients. In Figs. 7 and 8, the HTC trend in function of Peclet number is depicted, where the comparison between the different HTC correlations for tubes and bundles has been set, taking into consideration a fixed average value of LBE temperature.

4. Transients simulation

Different operational and accidental transients have been carried out with the aim to verify the capability of the new version of ATHLET system code and, at the same time, to assess whether the current spallation loop design can accommodate these transients.

Furthermore, a comparison with the results already obtained in a previous simulations campaign (Dierckx et al., 2009) using the RELAP5/mod 3.3 code has been carried out. This version of the code was previously applied in post-test analyses of experiments performed with CHEOPE, NACIE and CIRCE facilities, installed at the ENEA research center of Brasimone (Agostini et al., 2002; Coccoluto et al., 2009).

Below, all the tests foreseen in the current analysis campaign will be explained:

- **Flow rate variation:** the sensitivity of the layout with regard to small variations in flow rate needs to be assessed.
- **Main pump failure:** blockage or coast-down of the main pump will immediately affect the surface heights in the loop. Moreover, the cooling capabilities will be lost.
- **Beam off/beam on:** density changes of the LBE as a result of changes in beam intensity result in changes in the height of the additional free surface below the nozzle. The buffer space should be large enough to accommodate these changes without filling the complete buffer and keeping the necessary inlet pressure head for the main pump.
- **Loss of heat sink:** the loss of cooling flow through the secondary side of the spallation heat exchanger will have an immediate impact on temperatures and therefore densities. In turn, these will influence the free surface heights.

A test matrix of the simulations performed in the present study is shown in Table 4, which summarizes the adopted boundary conditions and the main variables that were monitored.

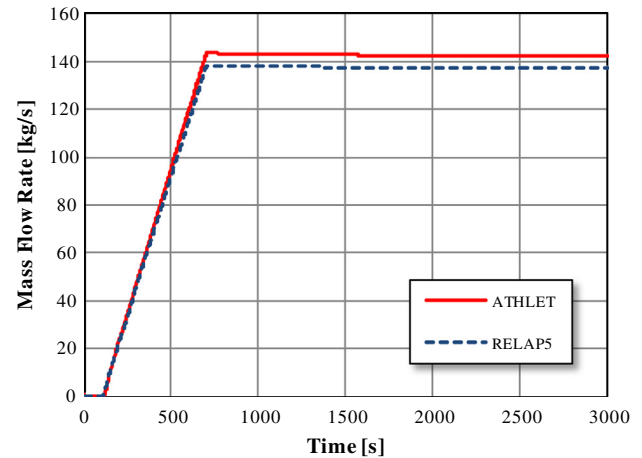


Fig. 9. LBE mass flow rate in the primary loop.

4.1. Start-up and steady state analysis

The main goal of this first simulation is to analyze the attainment of the operational conditions using a specific start-up procedure. It will be important to keep under control the liquid level variation in the main vessel and in the beam tube with the purpose of obtaining an expected configuration in the target zone, characterized by a liquid metal–argon interface.

The pump is kept switched off for the first 100 s to obtain a correct pressure distribution in the loop with stagnant fluid condition and then the pump speed is changed linearly from 0% to 100% over the next 600 s.

The LBE flow rate in the secondary side of the HX is increased in a linear way during the first 100 s of the transient starting from 0 to 133 kg/s and then is maintained constant. The generated thermal power, due to the spallation reaction, is released in the target zone through the adjacent wall. In particular, it is not delivered in the first 200 s of the simulation to avoid a too high temperature increase until the mass flow rate is relatively low. Following this phase, from 200 to 1100 s the thermal power is linearly increased up to the value of 1 MW and so maintained.

The calculated LBE mass flow rate in function of time is depicted in Fig. 9. It was obtained for both codes in the junction located at the lower part of the return line, underneath the target area. In accordance with the variation in time of the pump rotation velocity, the flow rate increases during the transient from 100 to 700 s in a linear way, reaching a nominal value of about 142 kg/s with ATHLET code and 138 kg/s with RELAP5. The small difference between

Table 4
Test matrix of the simulations.

Test Name	Description	Monitoring variables	Conditions
A	Start-up and steady state analysis in nominal conditions	• T_{in} and T_{out} in the primary and secondary sides of HX; • Liquid level in the beam line and in the main “service” vessel; • T_{in} and T_{out} in the target; • p_{in} and p_{out} in the target; • p_{in} and p_{out} through the main pump; • LBE flow rate through the main pump and the target.	Nominal steady state conditions: • thermal power: 1 MW • T_{init}: 300 °C • ΔT target: 50 °C
B	Small variations of the primary flow rate ($\pm 2\%$ of the nominal pump rotation velocity)	• T_{in} and T_{out} in the primary and secondary sides of HX; • Liquid level in the beam line and in the main “service” vessel; • T_{in} and T_{out} in the target; • LBE flow rate through the target.	This and the following accidental scenarios begin with the steady-state conditions obtained at the end of Test A
C	Main pump failure	(Same variables of Test B)“	
D	Fast variation in proton beam intensity (beam OFF/beam ON)	(Same variables of Test B)“	
E	Loss of heat sink	(Same variables of Test B)“	

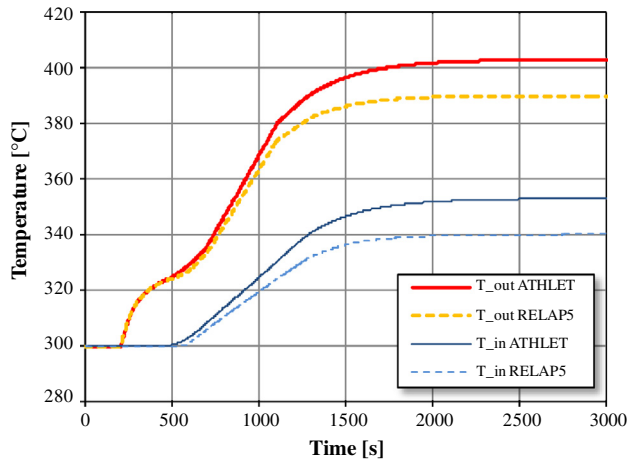


Fig. 10. Temperature through the target area.

the results obtained with the two codes is mainly imputable to the different temperature values reached during the transient. In Fig. 10 it can be observed the time trend of the LBE temperatures in the first and the last control volume of the target area, named TARGET_ZONE in the ATHLET nodalization and PIPE 150 regarding the RELAP5 one. Both temperatures remain constant in the first 200 s, due to the fact that in this first transient phase thermal power is not provided.

After 2500 s from the beginning of the calculation, the LBE temperature calculated with ATHLET code reaches an asymptotic value of 352 °C in the inlet section of the target and, approximately, 402 °C in the outlet section. On the other hand, the variations in time of the same parameters obtained with RELAP5 code show a similar trend, achieving, nevertheless, values of about 12 °C lower than the previous ones. Such effect is due to the different HTC correlations implemented inside the two system codes, as shown in Fig. 11.

The most remarkable difference can be noted for HTC evaluated inside the HX tubes, compared with that calculated for the HX tube bundles. Such behaviors during the transient were foreseeable if one takes into consideration the graphs in Figs. 7 and 8.

The last parameter that can be put in evidence is the liquid level in the main expansion vessel and in the lower part of the beam line. At the beginning of the transient calculation, both LBE free levels have the same elevation with respect to the lower point of the spallation circuit, which corresponds to a value of 5.1 m (see

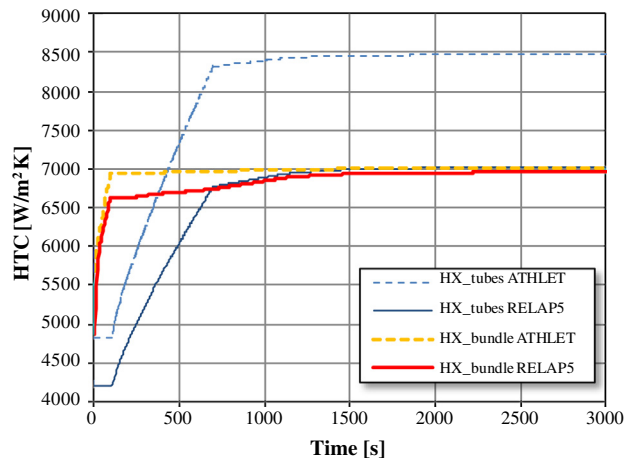


Fig. 11. HTCs calculated through the inner and outer sections of HX.

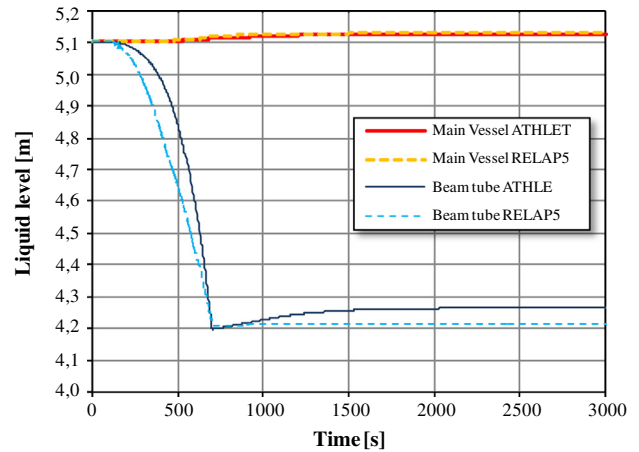


Fig. 12. Liquid level variation in main vessel and beam tube components.

Fig. 12). These values modify themselves only when the mass flow rate increases, due to the pump start-up, which occurs at 100 s.

The liquid level in beam tube decreases till reaching the lowest elevation value of 4.2 m at the instant in which the mass flow rate achieves its nominal value. Such elevation coincides roughly to the end part of the beam tube. In this phase, one notices that the effect of the temperature increase on the liquid level is less than the effect of the increase in flow rate. Once mass flow rate reaches its nominal value at about 700 s, the level variation is not influenced significantly by this parameter. In the second phase, when the steady state conditions are reached, there is a slight increase in the LBE level calculated with ATHLET code till about 1500 s due to the heat up of the loop. The final liquid elevation performed with ATHLET settles around a value of 4.26 m, while it reaches 4.21 m using RELAP5.

For the liquid level in the main expansion vessel, a positive variation during the second phase when the power increases can be observed. In this case, the LBE level undergoes a very slight variation caused by the large vessel size, compared with that of the beam tube. At the steady state, this elevation corresponds to a value of about 5.13 m for both codes.

From the previous results, it can be pointed out that the liquid level in the beam line depends strongly on the loop flow rate and only in part on the density variation due to temperature changes, while the level in the main expansion vessel is mainly affected by the heat up of the LBE.

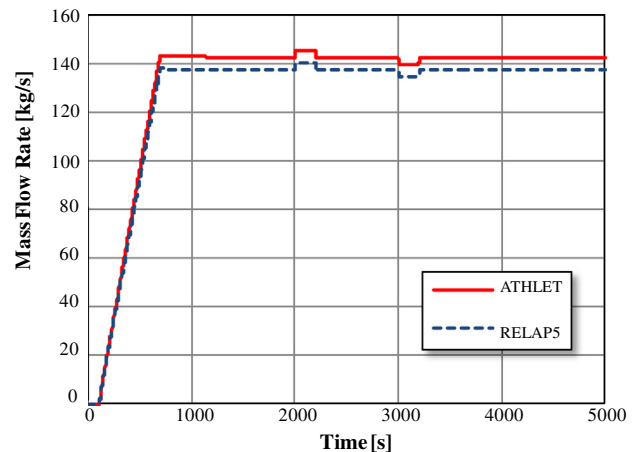


Fig. 13. LBE mass flow rate in the primary loop (Test B).

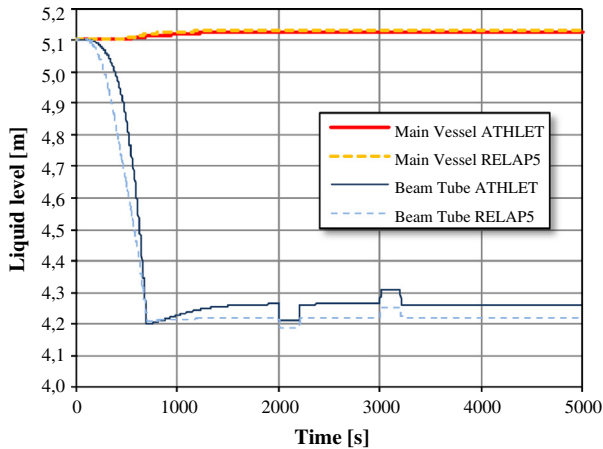


Fig. 14. Liquid level variation in main vessel and beam tube (Test B).

4.2. Accidental scenarios

The main goal of the first accidental transient is to analyze the behavior of the spallation loop in the event of small flow rate variation compared to the nominal flow rate. The required flow rate variation is obtained by changing the rotation pump velocity. The boundary conditions for the first 2000 s of the transient are the same as those from Test A. Subsequently, the pump rotation velocity is increased at first by 2% for a period of 200 s and, after another period of steady state, is reduced by 2% again for 200 s.

Fig. 13 shows the time trend of the LBE mass flow rate through the junction connecting the last two control volumes of the downcomer. It can be pointed out that the discrepancy in terms of total mass flow between the results achieved with both ATHLET and RELAP5 codes remain the same also during the variation of pump speed.

Finally, in Fig. 14, the effect of small changes in flow on the LBE level in the beam line and in main vessel is observed. When the flow increases, the liquid level in beam tube drops. Vice versa, during the phase in which the mass flow rate diminishes, it can be noted an augmentation of the LBE free surface elevation with respect to the nominal value. The LBE level variation in both situations is noticeable to have a greater extent by ATHLET simulation compared to the RELAP5 one. This can be attributed to the absence of the buffer surface in the ATHLET simulation, whereas in the RELAP5 simulation a buffer zone is present.

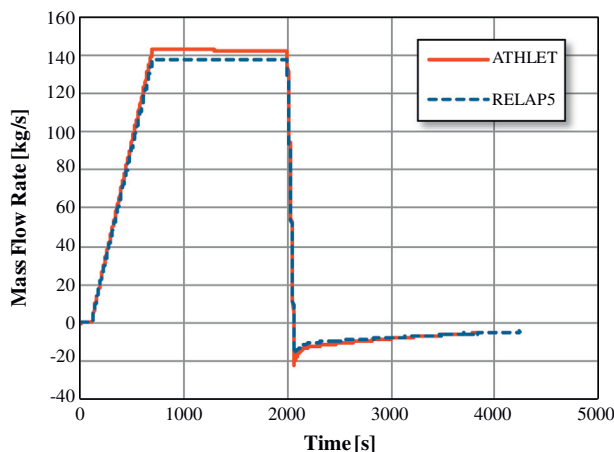


Fig. 15. LBE mass flow rate in the primary loop (Test C).

The liquid level in the service vessel undergoes a minor change compared to the one shown in the beam line. In fact, when the LBE mass flow rate increases or decreases, it remains stable caused by the large amount of LBE, which acts as a disturbance compensation.

The second accidental scenario is represented by Test C, in which the purpose is to analyze the behavior of the spallation circuit when a sudden failure of the main pump occurs. To obtain this particular condition, the pump rotation is reduced to zero within a few seconds, with the consequent reduction of the LBE mass flow rate. The pump is turned off at 2000 s with a total coast-down time of 60 s and then is kept off in order to evaluate changes in temperature of the liquid metal in the circuit, while the proton beam is still on, thus inducing heat in the spallation zone.

Fig. 15 shows the evolution in time of the LBE mass flow rate. The flow decreases abruptly after pump failure, and after about 80 s from pump blockage, the flow becomes a negative value. The inversion of the LBE circulation in the circuit is due to the presence of a considerable temperature difference between the hot and cold legs of the spallation loop causing a natural circulation force in reverse direction. After the sharp decrease, the mass flow rate tends to stabilize on a small negative value, even if the calculation stops at around 4000 s due to the high temperatures reached by the eutectic alloy. This behavior is closely related to the inability of establishing a strong natural circulation caused by the “wrong” positions of both heat source and the heat sink. To boost natural circulation of hot and cold fluids, the heat source must be at lower elevation than the heat sink.

The effect of decreasing mass flow on the LBE temperature in the target is shown in Fig. 16. As soon as the mass flow begins to decrease, the temperature quickly reaches a peak of 1280 °C. The time elapsed between the pump stop and the temperature peak turns out to be about 55 s. It means that the temperature peak occurs before the pump rotation velocity goes completely down, due to flow stagnation in the target. This sudden increase in temperature is very dangerous, because it requires urgent action to shut-down the proton beam in order to limit serious structural problems. In the instants following the peak, a drop in temperature can be noticed due to the fact that the target begins to get a fraction of cold liquid from the right side of the circuit, as a result of the flow reversal. After this, a continuous temperature increase in the target can be observed, since only small natural circulation establishes, as already mentioned above.

The third accidental scenario (Test D) regards the fast change in the proton beam intensity, to simulate a sudden loss/recover of heat in the loop. The heat generated by the spallation reaction de-

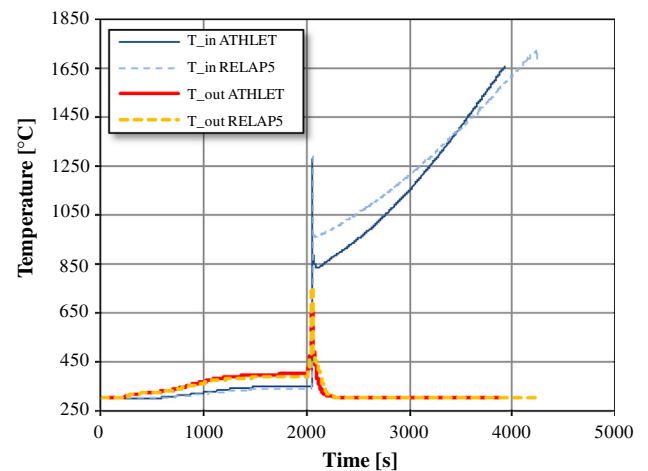


Fig. 16. Temperature through the target area (Test C).

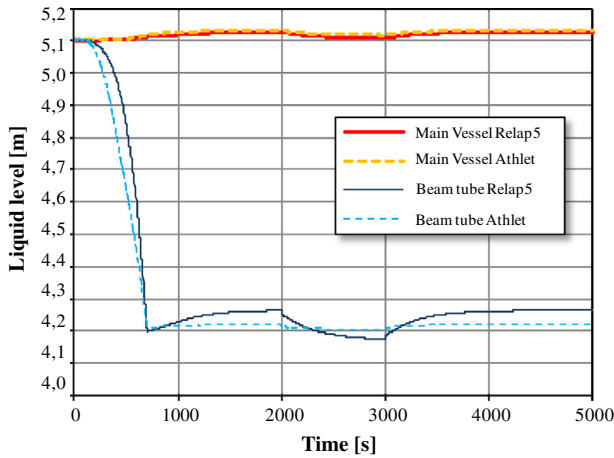


Fig. 17. Liquid level variation in main vessel and beam tube (Test D).

depends on the intensity of the proton beam and for this reason, the simulation is carried out considering the variation in thermal power directly to the target.

The thermal power in the target is put to zero at 2001 s with a sharp decrease of 1 s and then maintained equal to zero for another 1000 s. Subsequently, it is returned back to the nominal value between 3000 and 3001 s and finally remains at the nominal value of 1 MW in the remaining part of the analyzed transient.

In this test, the main variables monitored are the beam line and the service vessel's liquid levels, shown in Fig. 17. As expected, there is a decrease in both liquid levels from the instant in which thermal power is shutdown. The drop is more pronounced in the beam tube than in the service vessel due to the different dimensions of these two components.

The current behavior is essentially related to the small increase of LBE mass flow rate in the circuit. This augmentation derives in turn from a LBE density variation resulting from the absence of heat generation. The lack of thermal power supplied in the target structure causes a rapid lowering of temperature, which is noticeable in Fig. 18.

When thermal power comes back linearly to its nominal value at 3001 s, the level begins to rise, settling at the steady value. In terms of liquid elevation, the calculated variation in beam line during the accidental transient is around 8 cm for the ATHLET simulation and only 2 cm for the RELAP5 calculation. This again shows the importance and functionality of the buffer zone.

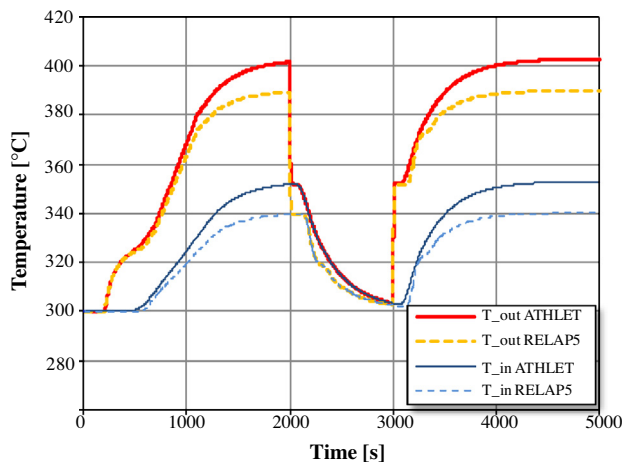


Fig. 18. Temperature through the target area (Test D).

The last accidental scenario concerns loss of heat sink (LOHS) in which the behavior of the entire loop is analyzed as soon as a sudden loss of cooling flow through the secondary side of the HX occurred. The way in which that condition is simulated consists in putting down rapidly the cooling flow through the secondary side of HX after reaching steady state conditions. For the ATHLET nodalization, it acts on the secondary side pump device, which represents a dummy element of the secondary loop nodalization, used only to simulate a constant mass flow rate. Regarding the RELAP5, the reduction of flow rate is obtained modifying the time trend of such parameter through a time-dependent junction (TMDPJUN 315 in Fig. 6).

After reaching a steady state, the cooling mass flow rate undergoes a linear reduction from 2000 to 2060 s and, successively, is left at zero for the entire duration of the transient. The LBE mass flow rate in the primary circuit, shown in Fig. 19, is affected by the lack of heat removal from the HX. When the accident occurs, the LBE mass flow begins to reduce, due to the increase in temperature. The loss of cooling in the HX causes a decrease in the LBE density and consequently an increase in the liquid levels in the hydrodynamic components with the presence of a free surface inside of them.

The increase in temperature upstream and downstream the heat exchanger is shown in Fig. 20. It can be noted that during

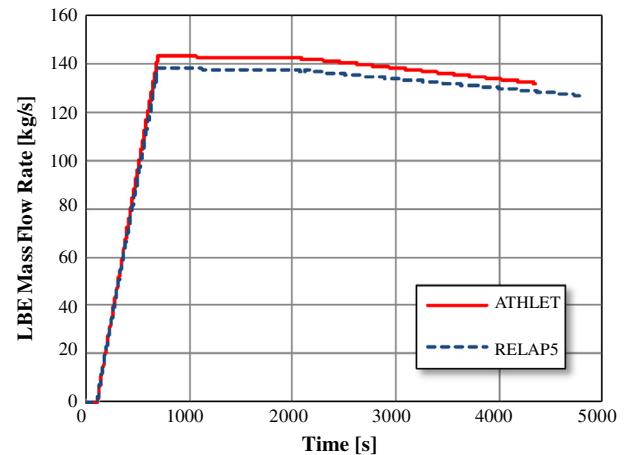


Fig. 19. LBE mass flow rate in the primary loop (Test E).

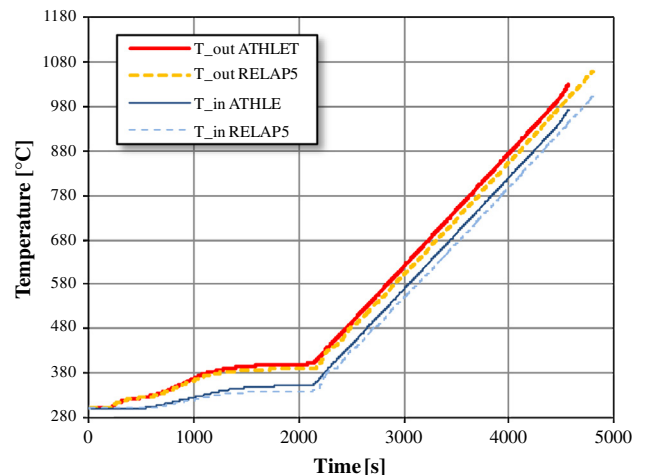


Fig. 20. Temperature upstream and downstream of the primary side of the HX (Test E).

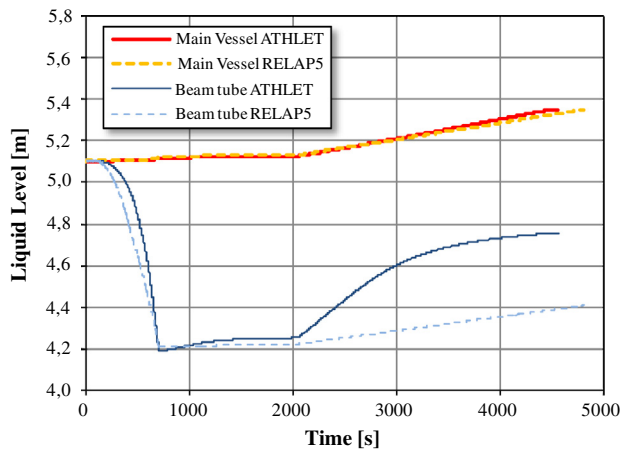


Fig. 21. Liquid level variation in main vessel and beam tube (Test E).

the period in which the loss of cooling in HX secondary side occurs (around 2060 s), the outlet temperature increases until it reaches the value in the inlet section. From this instant, the trends are almost overlapped and the two temperatures increase linearly until the end of the transient.

In Fig. 21, the liquid levels in the expansion vessel and in beam line can be observed, which increase in all the volumes due to thermal expansion of the liquid metal.

In the beam line, it can be noticed a different trend of the liquid level between the two codes. In ATHLET simulation, the level rises faster than in the “service” vessel during the first 1000 s after the secondary mass flow fall. This behavior is related principally to the different dimension of the two components, in the smaller of one, the thermal expansion is more noticeable in terms of liquid height. Furthermore, after 3000 s from the beginning of the transient, the suction effect due to forced circulation by junction connecting the beam tube with the target region prevents the beam line to be filled completely with LBE.

In RELAP5 calculation, the level variation of LBE is again more limited in the beam line compared to that obtained with ATHLET, due to the presence of the additional free surface (buffer region) in the nodalization, which permits to subdivide in a uniform manner the effect of thermal expansion through two components instead of just one.

5. Summary and conclusions

The purpose of the present paper is to give a brief overview on the new ADS system, taking into consideration the MYRRHA facility, developed at the SCK-CEN research centre with the aim to demonstrate principally the feasibility of transmutation of high-level nuclear waste. In particular, the attention has been focused on the reference design boundary conditions and general layout of the Spallation Loop foreseen in the MYRRHA/XT-ADS project, which represents a sort of interface device between the proton accelerator and the reactor core, allowing the dissipation of the heat generated by the spallation reaction.

A thermal-hydraulic analysis of such spallation loop during operational and accidental transients is carried out using for the first time the new version of the system code ATHLET, in which the physical properties of lead, lead–bismuth eutectic alloy (LBE) and sodium have been implemented. Moreover, the obtained results have been compared with the ones achieved using a modified version of the RELAP5/mod 3.3 system code, in which the lead and LBE properties have been implemented at University of Pisa.

The analysis of the start-up transient (Test A) shows that the liquid level in the beam line reduces almost completely as required to guarantee the right position of the liquid interface. Furthermore, the LBE mass flow rate calculated with both codes through the primary loop reaches similar nominal values in about 700 s. The Test A has also verified the correct removal of the heat generated in the target by the foreseen HX. Nevertheless, the different heat transfer correlations implemented in ATHLET and RELAP5 codes have played a crucial role in obtaining the final temperature distribution in the loop.

The most dangerous accident concerning the reached high temperature is the main pump failure without beam shut-off (Test C). LBE temperature in the target region has reached a value of about 1300 °C in few seconds. Consequently, the mass flow rate of the spallation loop and/or the rotation velocity of the main pump should be monitored accurately by an automatic control system in order to avoid problems related to the high temperatures obtained in the target region.

In the case of loss of heat sink (Test E), the mass flow rate in the primary loop decreases and the target temperature increases with a sufficiently slow rate compared to the case of the pump failure accident. From a safety point of view, the time constants involved in the dynamics of the transient allow the automatic control system or operators to have a sufficient margin of time to switch off the proton beam.

However, the simulation of the spallation area with the target free surface, the free falling jet and the buffer space was not possible in a straightforward manner within both codes. As it can be noticed in the nodalizations of Figs. 5 and 6, a workaround was arranged using junctions between different pipe components representing the beam tube and the target area and pressure loss coefficients were adapted. Moreover, the fixed height of the beam tube object implemented in the ATHLET nodalization has affected significantly the LBE level elevation. Namely, for a given height of this pipe, the pressure value that established above the free surface has influenced the position of the LBE level. This resulted in a working model for the spallation loop, but the correctness of the approach for the exact simulation of the free surface levels still needs additional investigation.

A further step will be to take advantage of the coupling technique already available between the ATHLET system code and the computational fluid-dynamic code ANSYS-CFX in order to proceed to a very detailed thermal-hydraulic analysis of the whole loop (Papukchiev et al., 2009). In particular, it will be easier with such method to observe the variation of the two free surfaces located inside the target area during incidental transients and to understand the impact that these changes have on the main parameters of the entire circuit.

References

- Abderrahim, H.A. et al., 2011. The MYRRHA ADS programme in Belgium: a multinational demonstration programme for incineration of spent nuclear fuel wastes. In: Technology and components of Accelerator Driven Systems, ISBN 978-92-64-11727-3, pp. 33–42, OECD 2011.
- Agostini, P. et al., 2002. Natural circulation of lead-bismuth in a one-dimensional loop: experiments and code predictions. In: Proceeding of the 10th International Conference on Nuclear Engineering (ICONE-10), Arlington, Virginia, USA.
- ATHLET User's Manual, 2009. ATHLET/Mod.2.2 Cycle A, ©GRS(mbH).
- Batta A., Class A., 2009. Hydraulic analysis of the heavy liquid metal free surface in the windowless XT-ADS Spallation target. In: Proceeding of the 13th International Topical Meeting on nuclear Reactor Thermal Hydraulics (NURETH-13), Japan, September 27–October 2, 2009.
- Bianchi, F. et al., 2001. Implementation and preliminary verification of the RELAP/ PARCS code for Pb–Bi cooled subcritical system, AccApp01 & ADITTA01, Nuclear Application in the new Millennium, Reno, Nevada, USA.
- Cheng, X., Tak, N., 2006. Investigation on turbulent heat transfer to lead–bismuth eutectic flows in circular tubes for nuclear applications. Nucl. Eng. Des. 236, 385–393.

- Class, A.G. et al., 2011. XT-ADS Windowless spallation target thermo-hydraulic design and experimental setup. *J. Nucl. Mater.* **415**, 378–384.
- Coccoluto, G. et al., 2009. Heavy liquid metal natural circulation in a one-dimensional loop. In: Proceeding of the 17th International Conference on Nuclear Engineering (ICONE-17), Brussels, Belgium.
- Dierckx, M. et al., 2009. MYRRHA spallation target loop: RELAP5 simulation. VELLA project, Deliverable 52 in the 6th Framework Program of the European Commission.
- Guertin, A. et al., 2011. XT-ADS windowless spallation target design and corresponding R&D topics. In: Technology and components of Accelerator Driven Systems, ISBN 978-92-64-11727-3, pp. 335–345, OECD 2011.
- Kirillov, P.L., Ushakov, P.A., 2001. Liquid-metal heat transfer in rod bundles. *Therm. Eng.* **48** (2), 127–133.
- Maes, D., 2006. Mechanical design of the small-scale experimental ADS: MYRRHA. *Energy Conv. Manage.* **47**, 2710–2723.
- Mansani, L. et al., 2011. The design of an experimental ADS facility (XT-ADS) and of a European industrial Transmutation Demonstrator (EFIT). In: Technology and Components of Accelerator Driven Systems, ISBN 978-92-64-11727-3, pp. 321–334, OECD 2011.
- Mikityuk, K., 2009. Heat transfer to liquid metal: review of data and correlations for tube bundles. *Nucl. Eng. Des.* **239**, 680–687.
- NEA/OECD, 2002. Accelerator-Driven Systems (ADS) and Fast Reactors (FR) in Advanced Nuclear Fuel Cycles. A Comparative Study. In: Proceeding of the Nuclear Energy Agency – Organization for Economic Co-operation and Development, ISBN 92-64-18482-1, Paris.
- OECD/NEA, 2007. Handbook on lead–bismuth eutectic alloy and lead properties, materials compatibility, thermal-hydraulics and technologies. OECD 2007, NEA No. 6195.
- Papukchiev, A. et al., 2009. Extension of the simulation capabilities of the 1D system code ATHLET by coupling with the 3D CFD software package ANSYS CFX. In: Proceeding of the 13th International Topical Meeting on nuclear Reactor Thermal Hydraulics (NURETH-13), September 27–October 2, 2009, Japan.
- Pfrang, W., Struwe, D., 2007. Assessment of correlations for heat transfer to the coolant for heavy liquid metal cooled core design. Wissenschaftliche Berichte Forschungszentrum Karlsruhe, FZKA 7352.
- Seban, R.A., Shimazaki, T.T., 1951. Heat transfer to a fluid flowing turbulently in a smooth pipe with walls at constant temperature. *Trans. ASME* **73**, 803–809.
- Stanculescu, A., 2000. Accelerator driven systems (ADS) and transmutation of nuclear waste: options and trends. Workshop on Nuclear data and nuclear reactors: physics, design and safety, April 2000, Trieste.